AI FITNESS COACH

MotionTech

DATE

2025.12.27

24/7 AI Personal Trainer

Real-time, Low-latency AI Fitness Coach

MotionTech Innovators

AI FITNESS COACH

video performance

DATE

2025.12.27

AI FITNESS COACH

video performance

DATE

2025.12.27

CONTENTS

catalogue

CONTENTS

01 Difficulties with personal training

02 Our Goal

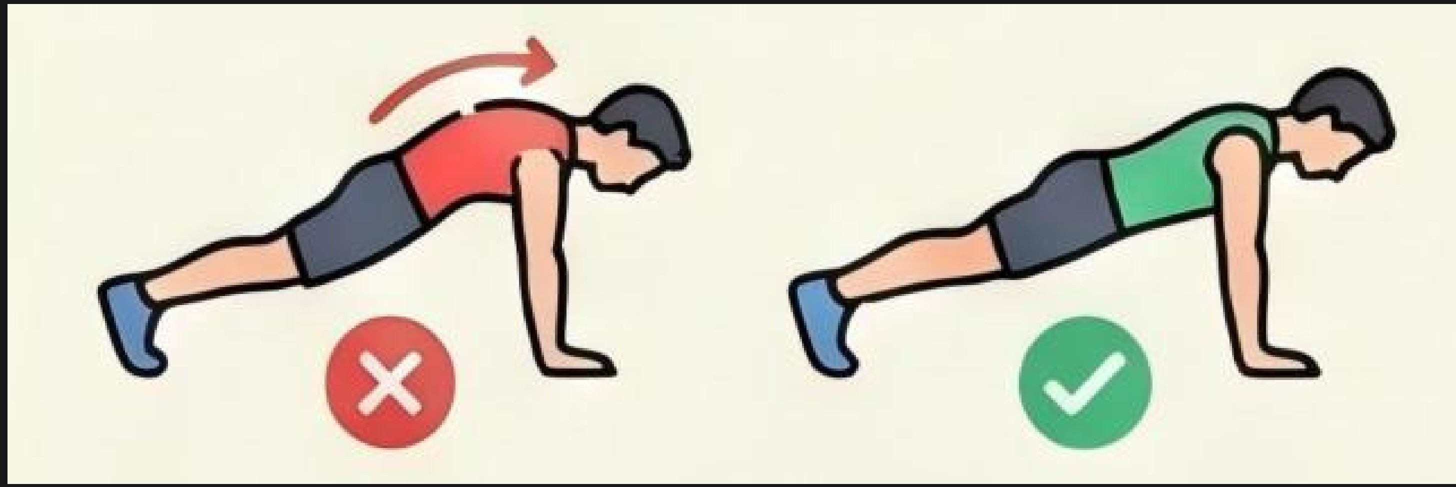
03 overall solution

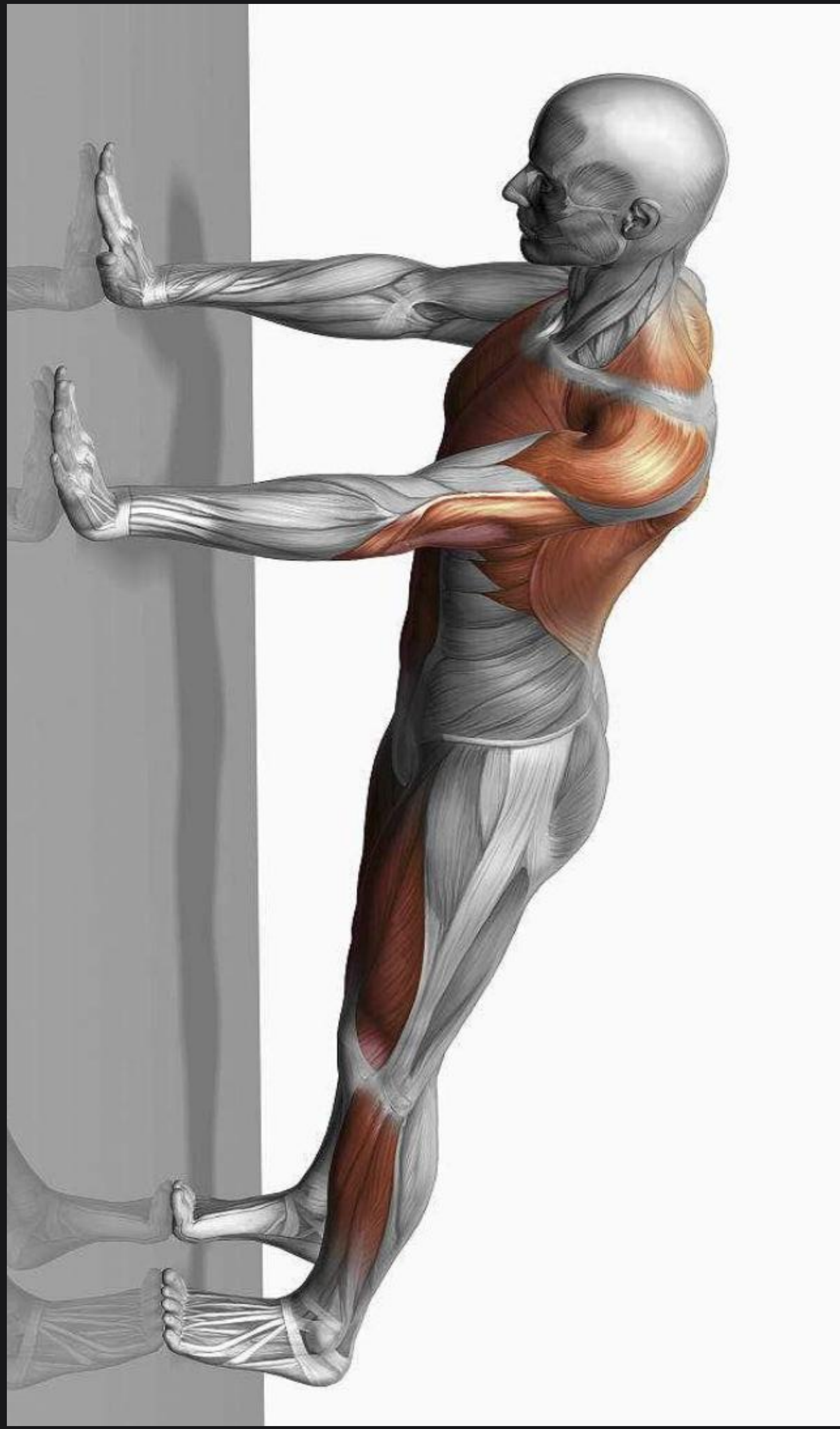
04 hardware module

05 Backend Module

06 Summary and Outlook

Difficulties with personal training





No Objective Feedback。

Errors Corrected Too Late。

Expensive & Inaccessible。

Our Goal

Primary Objective

Build a real-time, low-cost system that detects, corrects, and evaluates motion quality.

Real-time

Instant feedback during exercise

Objective

Quantitative motion evaluation

Affordable

Cost-effective solution

Home Fitness

24/7 accessible training

Overall Solution

Component 01
AI Vision

Function

Sees Posture

- Real-time pose estimation via camera feed
- Skeleton tracking & joint angle calculation
- Visual feedback & error highlighting

Component 02
Hardware Sensors

Function

Feels Motion

- IMU sensors capture motion dynamics
- Accelerometer & gyroscope data fusion
- Physical feedback via haptic alerts

Component 03
Cloud AI

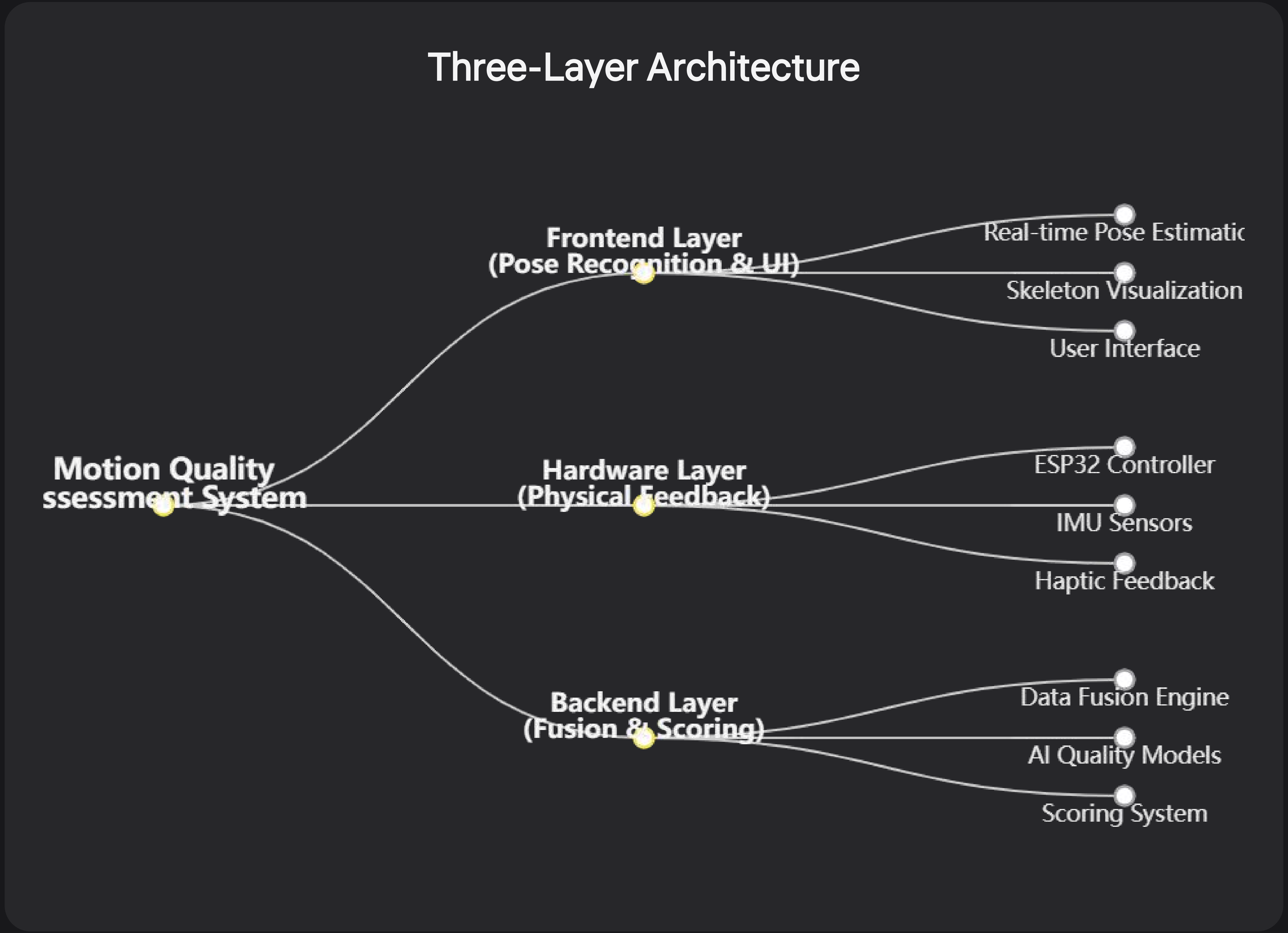
Function

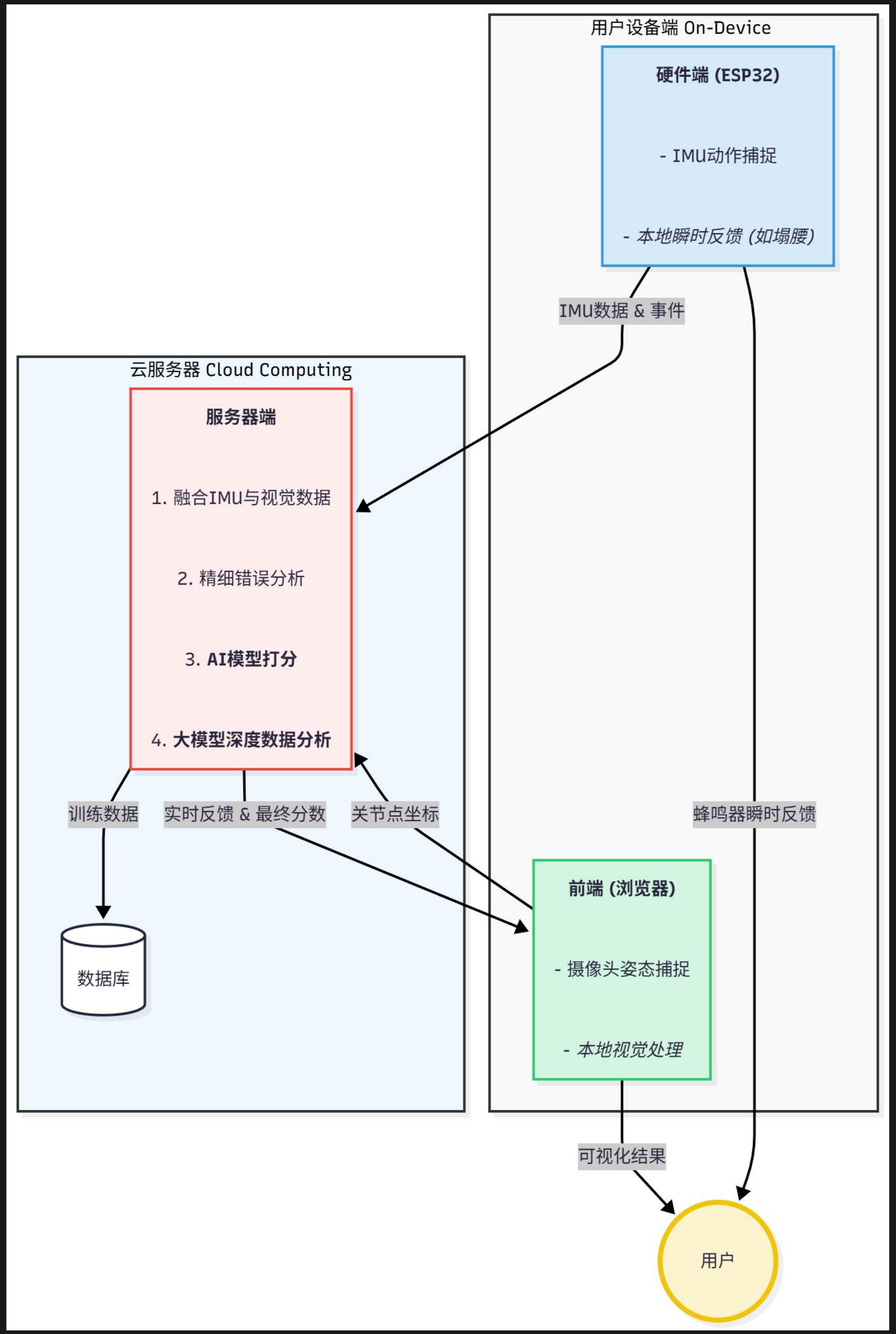
Thinks & Evaluates

- Deep learning models for quality assessment
- Multimodal data fusion & analysis
- Personalized coaching insights

A 24/7 AI personal trainer that can **see**, **feel**, and **think**.

System Architecture





Why Multimodal Matters

Vision Alone

Blind Spots

Cannot detect motion when body parts are occluded or outside camera view

Limited Depth

2D video lacks 3D motion dynamics and acceleration data

Sensors Alone

Lack Context

No visual context of overall body posture and alignment

Placement Dependency

Accuracy depends heavily on precise sensor placement

Multimodal

Complete Understanding

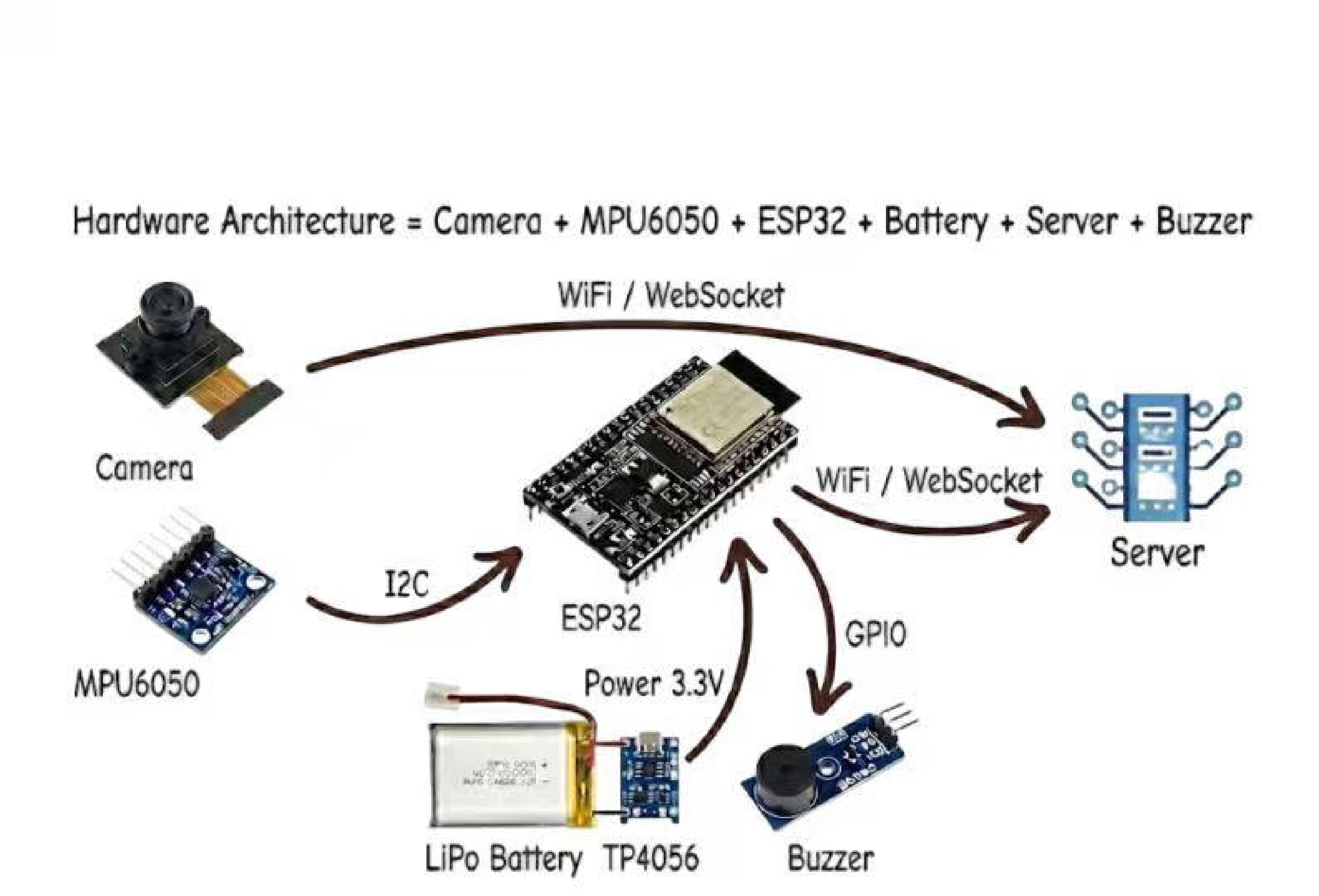
Combines visual data with motion dynamics for full context

Redundant Validation

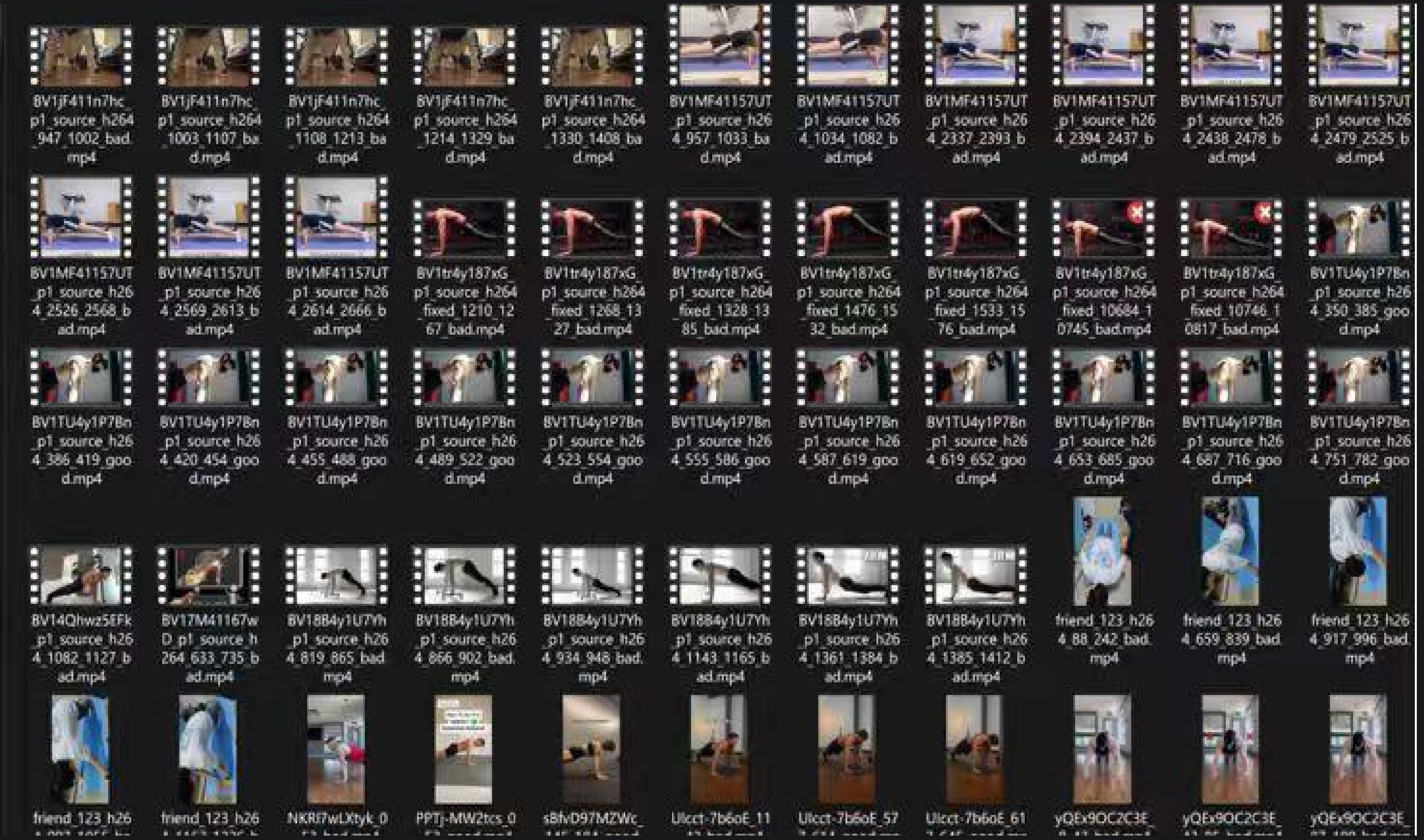
Cross-validates between modalities for higher accuracy

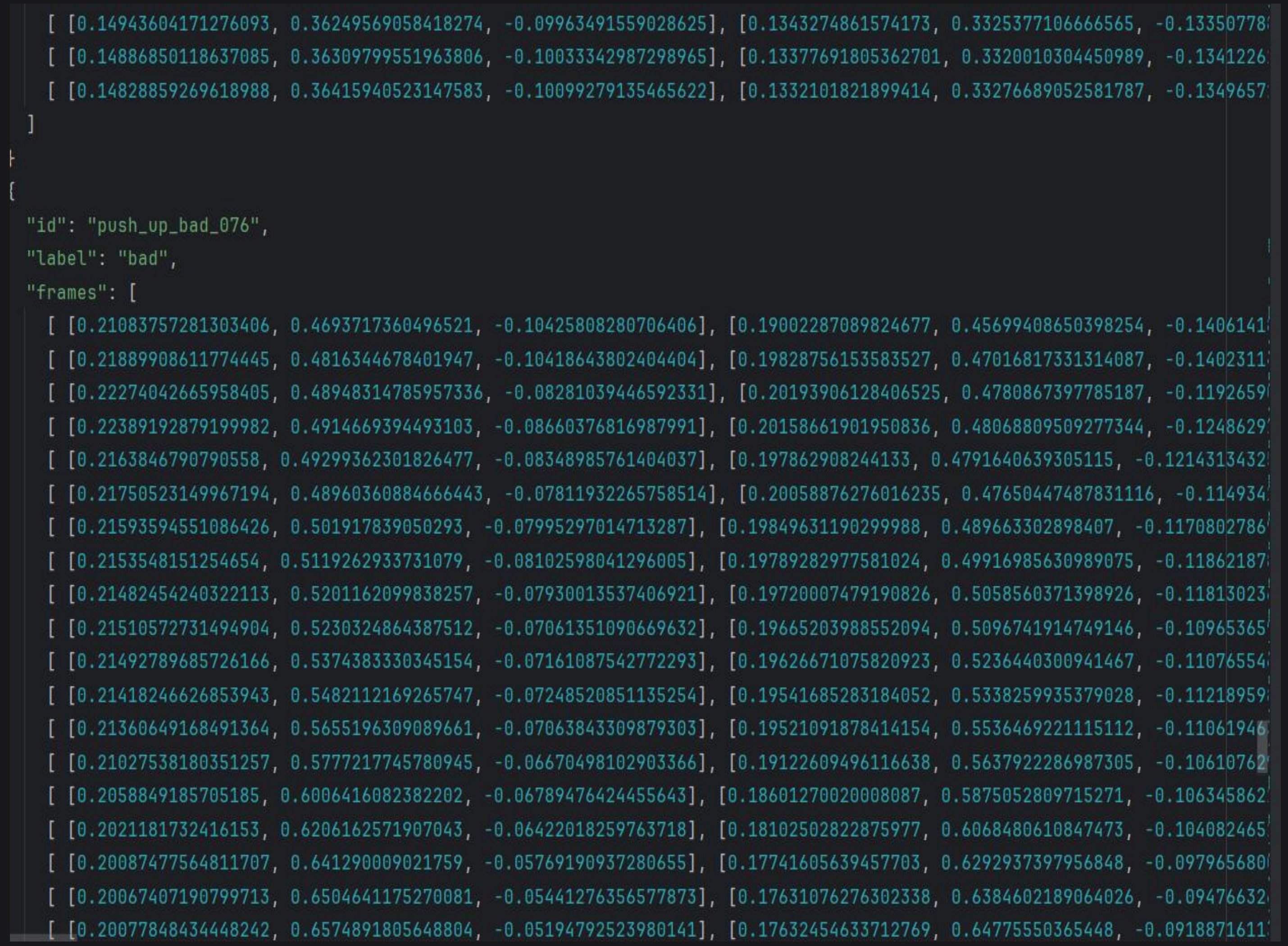
One modality is not enough.

Hardware Module



Self-bulid dataset





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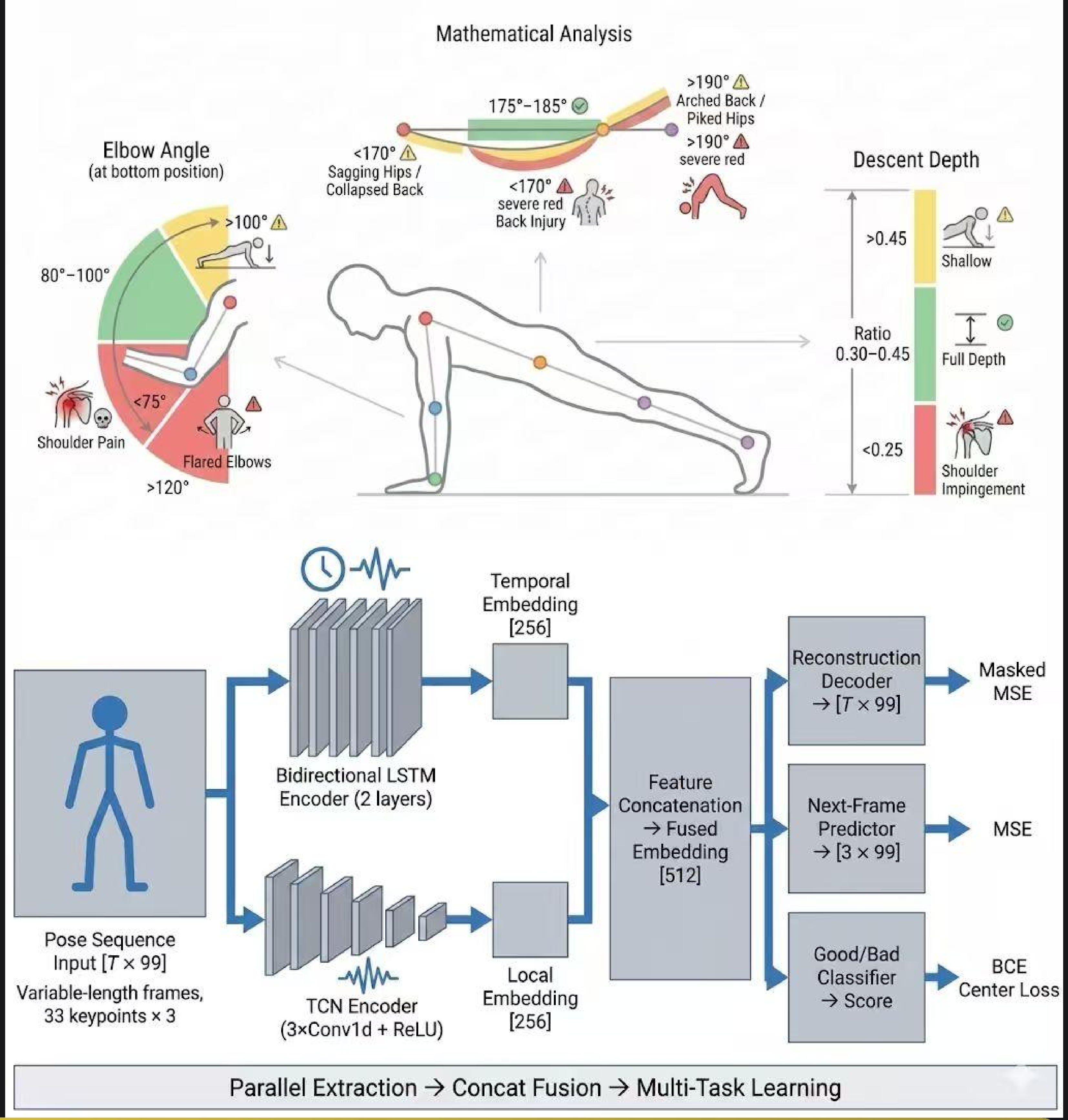
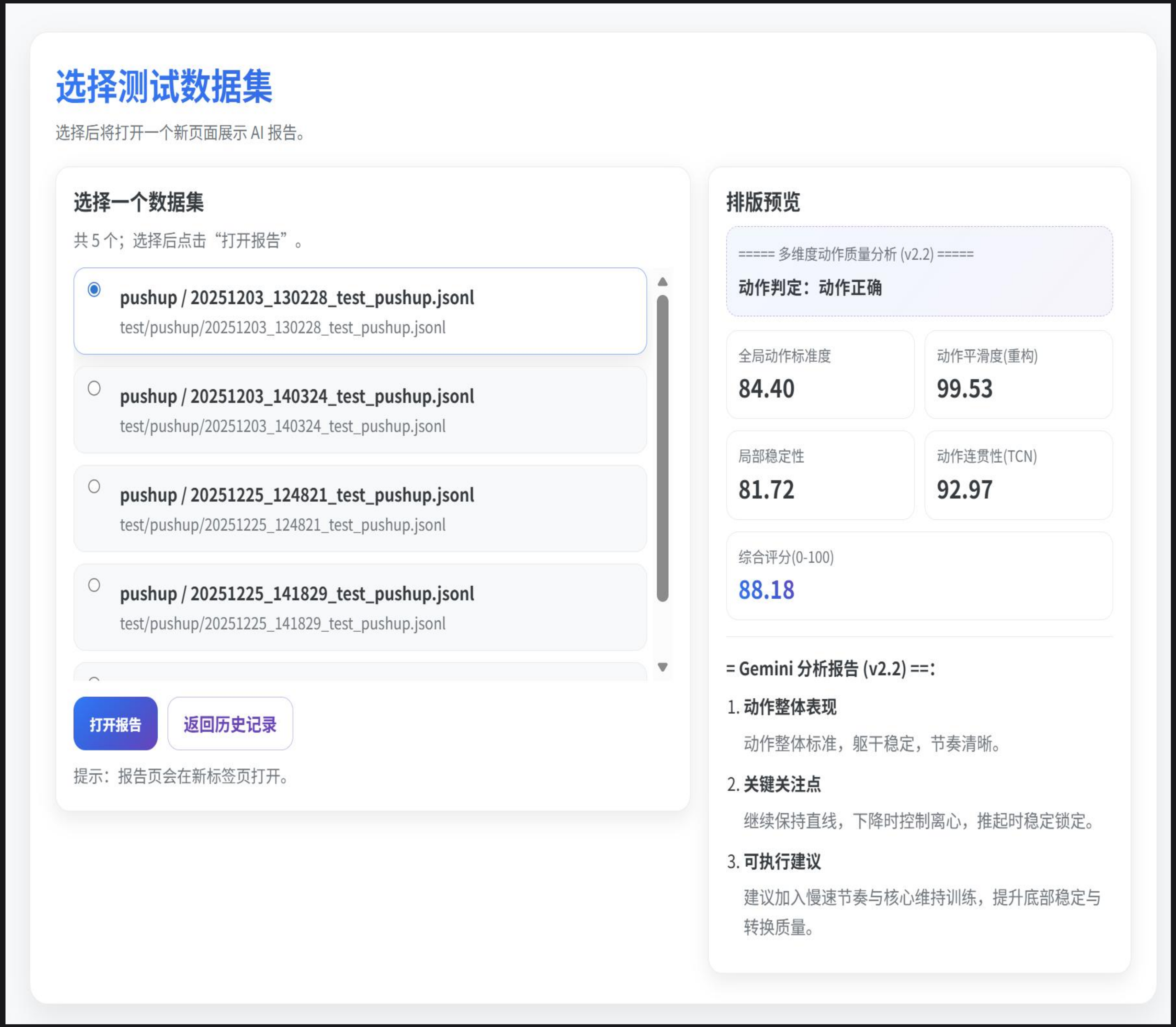
Mathematical threshold method for judging obvious problems

Fusing LSTM and TCN models for overall evaluation



We evaluate **quality**, not action type.

Final report



11 Final AI report

AI 智能分析报告

范围：数据集: 20251203_130228_test_pushup.jsonl

生成时间：2025-12-26 13:28:18 用户：test

结论概览

- 数据集：20251203_130228_test_pushup.jsonl
- 判定：动作整体标准，节奏稳定，核心控制良好

评分面板（本次测试）

- 全局动作标准度：84.40
- 动作平滑度(重构)：99.53
- 局部稳定性：81.72
- 动作连贯性(TCN)：92.97
- 综合评分(0-100)：88.18

1) 动作整体表现

您的俯卧撑综合评分 **88.18**，整体姿态较标准、稳定性表现良好。 不过从分项对比看，**动作连贯性(TCN) 92.97** 虽然不低，但相较 **动作平滑度(重构) 99.53** 更容易成为“短板项”，建议把重点放在推起阶段的发力路径与节奏一致性上。

2) 动作阶段解读

- **下降阶段**：离心控制更强，整体更稳定。
- **底部稳定与顶部锁定**：总体可控，但局部稳定性 **81.72** 提示底部与转换阶段仍有轻微波动空间。
- **推起阶段（主要优化点）**：存在发力路径不够一致的问题，影响连贯性。

补充观察（用于快速提升质量）：

- 手肘外扩角度约 **53°**（偏大），建议 **收肘**，让肘部轨迹更贴近躯干，减少肩部压力并提升推起效率。
- 俯卧撑深度角度约 **132°**（偏浅），建议将最低点控制到 **100°以下**，以获得更充分的训练刺激。

动作质量拆解

1. **身体线条**：头-肩-髋-踝整体保持接近一条直线；下降与推起过程中躯干稳定。
2. **深度与幅度**：下降幅度充分，推起阶段锁定清晰；未见明显代偿。
3. **节奏与控制**：动作平滑，离心（下降）控制更强，推起发力集中。

训练建议（用于继续提升）

- **慢速俯卧撑 3-1-3**：3 秒下降、底部停 1 秒、3 秒推起，强化离心与底部稳定。
- **肩胛控制**：肩胛骨俯卧撑 2 组 × 12 次，提升肩胛带稳定。
- **核心维持**：平板支撑 2 组 × 45–60 秒，保持中立位。

> 动作标准，可以稍微增加俯卧撑深度。

生成时间：2025-12-26 13:28:41 用户：test

结论概览

- 数据集：20251203_140324_test_pushup.jsonl
- 判定：存在**塌腰（核心失稳/腰椎过伸倾向）**，下降阶段更明显

评分面板（本次测试）

- 全局动作标准度：84.40
- 动作平滑度(重构)：99.53
- 局部稳定性：81.72
- 动作连贯性(TCN)：92.97
- 综合评分(0-100)：88.18

1) 动作整体表现

本次综合评分 **88.18**，动作平滑度 **99.53** 表现很强，但塌腰会让“好看的连贯性”在关键阶段掉分。 从结果表现看，问题更集中在 **核心抗伸展不足** 导致的躯干线条波动，进而影响推起阶段的发力稳定与连贯性。

2) 动作阶段解读

- **下降阶段**：更容易出现腰部下沉，导致动作线条被“折断”。
- **底部稳定（重点）**：局部稳定性 **81.72** 提示底部转换阶段的支撑与核心张力不足。
- **推起阶段**：当核心松掉时，推起会变成“上肢硬推”，路径不稳，连贯性受影响。

补充观察（用于快速提升质量）：

- 手肘外扩角度约 **53°**（偏大），建议 **收肘**，让肩胛带更稳定。
- 俯卧撑深度角度约 **132°**（偏浅），建议先在“不塌腰”的前提下逐步做到 **100°以下**。

关键问题定位

1. **核心抗伸展不足**：下降时腹部张力不足，导致腰椎代偿（看起来像腰部下沉）。
2. **骨盆控制不足**：髋部稳定性不够，身体线条无法持续保持直线。
3. **肩胛带补偿**：在底部转换阶段，肩胛稳定性略弱，影响整体稳定。

立即可执行的纠正要点

- **口令**："收肋骨、收腹、夹臀"，让躯干保持中立位。
- **下降策略**：下降时不要“放松腰”，保持腹部持续用力。
- **底部停顿**：底部停 0.5–1 秒（不塌腰前提下），再推起。

训练建议（3 天一轮）

- **死虫 (Dead Bug)**：2–3 组 × 10 次/侧（核心抗伸展）。
- **平板支撑**：2 组 × 30–45 秒（保持骨盆中立）。
- **慢速俯卧撑 2-1-2**：先质量后数量。

> 保持腰腹绷紧，不要塌腰。

11 paper

International Journal of
EXERCISE SCIENCE

AI识图

Original Research

Development of a Standard Push-up Scale for College-Aged Females

MELANIE M. ADAMS‡[1], SOPHIE A. HATCH*[1], ELIZABETH G. WINSOR*[1], and CAITLYN PARMELEE‡[2]

[1]Department of Human Performance & Movement Sciences, Keene State College, Keene, NH, USA; [2]Department of Mathematics, Keene State College, Keene, NH, USA

*Denotes undergraduate student author, ‡Denotes professional author

ABSTRACT

International Journal of Exercise Science 15(4): 820-833, 2022. The ACSM/CESP push-up test exemplifies the limiting nature of the gender binary in fitness. Males perform the standard push-up (from toes) while females perform the modified push-up (from knees), even if capable of multiple standard push-ups. Differences in upper body strength are used to justify the test protocol. Though the load difference between modified and standard positions is substantially less than the gender strength gap. Additionally, current fitness ratings are over 30 years old. The purpose of this study was to develop a new standard push-up rating scale for college-age females. Cis-female college students (n = 72) were recruited to perform maximal repetitions in the modified and standard positions. Health history and physical activity information was gathered prior to the test. Trained research assistants provided standardized warm-up, modelled correct form, and administered the tests. Order of the tests was randomized and there was at least 48 hours between test days. Mean push-ups in the standard position was 9 (8.87) and 17.5 (11.76) in the modified position. Participants who resistance train did significantly more repetitions of each. Linear regression was used to develop an equation to predict standard push-up repetitions from modified repetitions. The equation was applied to the current repetition ranges for each fitness category, and a new standard scale was developed. The new scale ratings are similar to the Revised Push-up but lower than the Fitnessgram® Healthy Zone. The modified or "girl" push-up contributes to gender stereotypes about muscular fitness. Providing females with the option to be graded on the standard push-up is a step to reducing gender bias in fitness. Future research is needed to validate this scale.

KEY WORDS: Push-up test, gender, fitness testing, stereotype

Original Article

Biomechanical Evaluation of the Push-Up Exercise of the Upper Extremities from Various Starting Points

TOPALIDOU ANASTASIA[1,2], DAFOPOULOU GEORGINA[3], KLEPKOU EIRINI - EROFILI[4], AGGELIGAKIS JOHN[1], BEKRIS EVAGGELOS[5], SOTIROPOULOS ARISTOMENIS[5]

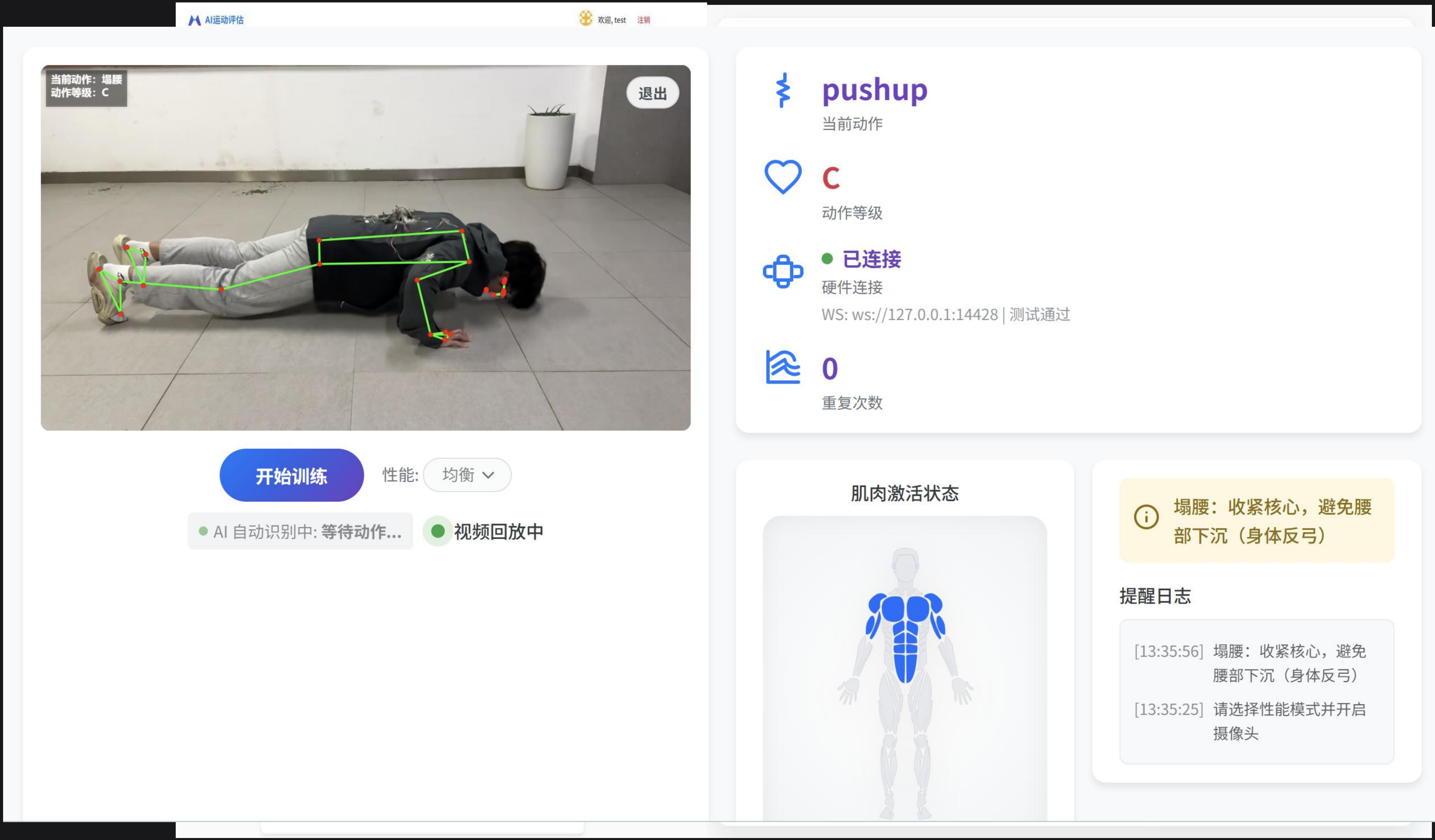
[1]*University of Grete, Faculty of Medicine, Department of Orthopaedic – Traumatology,GREECE*
[2]*Alexander S. Onassis Public Benefit Foundation, GREECE*
[3]*Aristotle University of Thessaloniki, Department of Physical Education and Sport Science, GREECE*
[4]*Technological Educational Institute of Thessaloniki, Department of Health and Medical Care, GREECE*
[4]*National and Kapodistrian University of Athens, Department of Physical Education and Sport Science, GREECE*

Published online: March 31, 2012
(Accepted for publication March 18, 2012)

Abstract:
The purpose of the present research was to evaluate the push-up exercise of the upper extremities in respect of biomechanics, to compare the muscle function, while changing the position of performance and to examine the torso's inclination during the exercise. The result is that the activation of the muscles, apart from the triceps brachii muscle, does not differentiate significantly in any of the positions. Moreover, the elevating the hands above the feet position is not recommended in protocols where the aim is to improve the muscle force because it displays the lowest mean value of vertical force and a low RFD. On the other hand, the standard push-up position is considered to be the most appropriate when the aim is to improve the triceps brachii muscle's force because it displays the highest RFD and the highest activation of this muscle. Finally, the correct body position during this exercise prevents from incorrect and damaging inclinations of the torso.

Key words: Biomechanical Evaluation , Electromyography , Push-Up Exercise.

GOALS OF THE CLIENT. athletes in sports such as baseball (10), training devices, and specially designed


AI运动评估
欢迎, test
注销
当前动作：塌腰
动作等级：C
退出
开始训练
性能:
均衡
AI 自动识别中: 等待动作...
视频回放中
pushup
当前动作
C
动作等级
已连接
硬件连接
WS: ws://127.0.0.1:14428 | 测试通过
0
重复次数
肌肉激活状态
塌腰：收紧核心，避免腰部下沉（身体反弓）
提醒日志
[13:35:56] 塌腰：收紧核心，避免腰部下沉（身体反弓）
[13:35:25] 请选择性能模式并开启摄像头

Team member roles

Modular division of labor and complementing each others strengths

1. **硬件端**

张哲威（ESP32硬件系统的构建，开发ESP32端固件程序，硬件端与服务器的WebSocket通信开发，硬件系统的供电方案设计），

2. **前端**

马亦麟（服务器搭建维护备案，所有网页页面的html与css、网页首页、登录注册、管理员控制、数据库管理的功能实现，姿态关节识别，数据可视化，数据csv导出，移动端迁移APP）。

3. **后端**

金慧婷（负责多源动作数据的获取与清洗，主导时序片段切割、关键帧抽取与格式统一，构建可直接用于 LSTM/TCN 训练的数据集体系，完善噪声过滤与数据增强流程，并协同模型端保持数据标准一致性）

陈依睿 （负责后端核心评估引擎的构建，主导基于骨骼关键点的实时几何规则诊断系统，实现俯卧撑、深蹲等动作的角度计算、质量评分与实时反馈，并设计多模态数据融合逻辑，协同硬件与前端完成系统闭环）

周琮越(主导各动作专属LSTM和TCN的融合模型的全周期开发与调试，设计多维度自监督误差策略构建“误差 - 评分” 映射体系，大模型API接入与调试，共同协调服务器系统)。

Summary & Outlook

What We've Accomplished

End-to-End Working System

Successfully integrated vision, hardware, and cloud AI into a cohesive motion assessment platform with real-world validation

Real-Time Feedback Loop

Achieved <50ms latency for instantaneous motion correction, proving the feasibility of AI fitness coaching

Scalable Architecture

Designed modular system supporting multiple exercise types and future extensibility to broader fitness scenarios

Future Directions

01 More Exercises

Expand beyond push-ups to squats, planks, lunges, and complex multi-joint movements

02 Mobile App

Develop iOS/Android applications for mainstream adoption and social fitness features

03 Finer-Grained Analysis

Implement joint-specific scoring and muscle group activation analysis